

Angular Goal Analysis and Robot Heading Error

Julio Quejido Barrera
Humanoide Robotik
Julio's Laboratories

July 28, 2026

Abstract

This project extends the geometric goal analysis developed in the previous projects by introducing an angular representation of the relative orientation between a mobile robot and multiple goals. For each candidate goal, the absolute direction of the corresponding robot-to-goal vector is calculated using the two-argument arctangent function atan2 . The difference between this goal angle and the robot orientation produces an angular error. Because angular quantities are periodic, the raw error is normalized to obtain the shortest signed rotational difference within the interval $[-\pi, \pi)$. The sign of the normalized angular error is then used to determine whether the robot should rotate left, rotate right, or continue moving straight.

1 Introduction

A mobile robot may operate in an environment containing several possible target positions. In this situation, selecting a goal cannot depend only on the distance between the robot and each candidate. The robot must also consider its current orientation, because a nearby goal located behind the robot may be less relevant than a goal positioned within its forward field.

This project introduces a robot heading vector and uses it to evaluate the relative position of multiple candidate goals. Each goal is processed individually, classified according to its position with respect to the robot heading, and tested using a visibility condition.

The geometric operations developed in Project 1 are reused to obtain the distance and normalized direction associated with each candidate goal. The new contribution of this project is therefore the combination of robot orientation, scalar-product analysis, spatial classification and visibility filtering.

2 Theoretical Background

2.1 Angular Representation of a Vector

For goal i , let $\mathbf{v}_{RG_i}$ denote the robot-to-goal vector. A two-dimensional direction vector can also be represented

by an angle measured from the positive global x -axis. The absolute angle toward goal i is

$$\theta_{G_i} = \text{atan2}(v_{y_i}, v_{x_i}). \quad (1)$$

The value θ_{G_i} describes the orientation of the robot-to-goal vector in the global coordinate system.

The two-argument arctangent is preferred over

$$\arctan\left(\frac{v_{y_i}}{v_{x_i}}\right) \quad (2)$$

because the one-argument expression cannot uniquely determine the quadrant of the vector and becomes undefined when $v_{x_i} = 0$. In contrast, atan2 considers the signs of both vector components and returns the correct angular quadrant.

Table 1: Comparison Between $\arctan(y/x)$ and $\text{atan2}(y, x)$.

Vec. (x, y)	Quad.	$\arctan(y/x)$	$\text{atan2}(y, x)$	Correct
$(3, 3)$	I	45°	45°	Yes
$(-3, -3)$	III	45°	-135°	No

2.2 Absolute Goal Angle and Robot Orientation

The robot orientation is represented by θ_R , while the absolute direction toward goal i is represented by θ_{G_i} . Both angles are measured from the same positive global x -axis. The robot orientation describes the direction in which the robot is currently facing, whereas the goal angle describes the global direction from the robot position toward the selected target. These quantities are absolute angles. The goal angle alone does not indicate how much the robot must rotate, because the current robot orientation must also be considered. A relative angular difference must therefore be calculated.

2.3 Angular Error

The raw angular error for goal i is

$$e_{\theta_i} = \theta_{G_i} - \theta_R. \quad (3)$$

This value represents the signed rotational difference between the current robot orientation and the direction toward the goal.

Before normalization,

$$e_{\theta_i} > 0$$

indicates a counterclockwise angular difference, while

$$e_{\theta_i} < 0$$

indicates a clockwise angular difference. When

$$e_{\theta_i} = 0,$$

the robot orientation is aligned with the goal direction.

The magnitude

$$|e_{\theta_i}|$$

indicates the size of the angular difference. However, direct angle subtraction does not always provide the shortest physical rotation because angular quantities are periodic.

2.4 Periodicity of Angles

Angular quantities repeat after every complete revolution. Therefore,

$$\theta = \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

Angles that differ by an integer multiple of 2π represent the same physical orientation.

For example, consider

$$\theta_R = -179^\circ$$

and

$$\theta_G = 179^\circ.$$

Direct subtraction produces

$$e_\theta = 179^\circ - (-179^\circ) = 358^\circ.$$

However, the shortest rotation from the robot orientation to the goal direction is

$$-2^\circ.$$

The large raw error appears because the two angles lie on opposite sides of the $-\pi/\pi$ boundary. This discontinuity shows that direct subtraction alone is not sufficient for obtaining the shortest rotational difference.

2.5 Angular Error Normalization

Because angular quantities are periodic, the raw angular error may not represent the shortest physical rotation between the robot orientation and the goal direction. The project therefore normalizes the error using

$$e_{\theta_i}^{\text{norm}} = (e_{\theta_i} + \pi) \bmod (2\pi) - \pi. \quad (4)$$

This operation maps any angular error to the interval

$$-\pi \leq e_{\theta_i}^{\text{norm}} < \pi.$$

The normalized value represents the shortest signed angular difference between the current robot orientation and the direction toward goal i .

When

$$e_{\theta_i}^{\text{norm}} \approx 0,$$

the robot is approximately aligned with the goal direction.

When

$$|e_{\theta_i}^{\text{norm}}| \approx \pi,$$

the goal direction is approximately opposite to the robot orientation. Because the normalization interval is $[-\pi, \pi)$, an angular difference of exactly π is represented as $-\pi$.

Small positive or negative values indicate that only a small rotation is required. The sign determines the direction of the shortest rotation, while the magnitude indicates its size. This normalization is essential in robotics because orientation variables wrap around after every complete revolution.

2.6 Rotation Direction Decision

The normalized angular error is converted into a simple motion decision:

$$\begin{aligned} e_{\theta_i}^{\text{norm}} < 0 &\Rightarrow \text{turn right,} \\ e_{\theta_i}^{\text{norm}} = 0 &\Rightarrow \text{go straight,} \\ e_{\theta_i}^{\text{norm}} > 0 &\Rightarrow \text{turn left.} \end{aligned}$$

The sign determines the shortest rotation direction, while the magnitude indicates how large the required rotation is.

3 Project Task

Define one robot position, a robot orientation angle, and several candidate goal positions as NumPy arrays. For every goal, compute the absolute direction using `atan2`, calculate the angular error between the goal direction and the robot orientation, and normalize the error to the interval $[-\pi, \pi)$. Convert the angular values to degrees

for interpretation and use the sign of the normalized error to determine whether the robot should turn left, turn right, or continue straight.